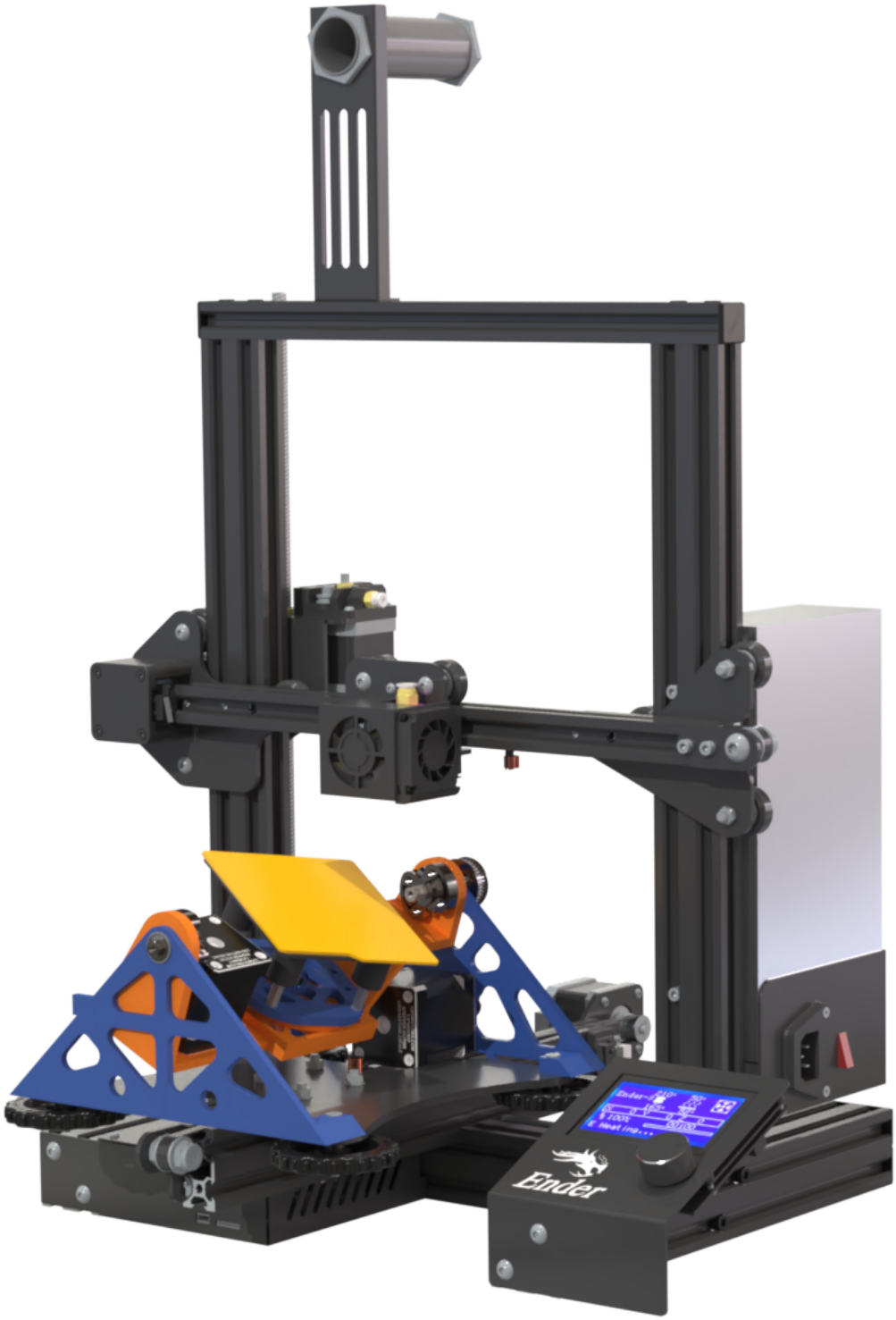
PENTOS

Assembly manual

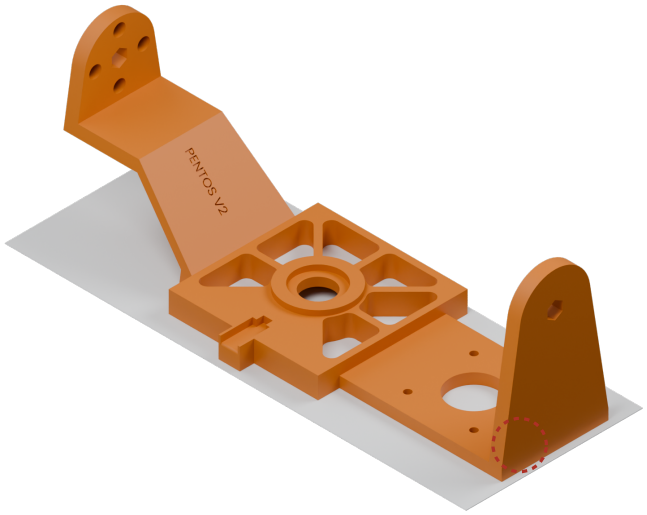


00 Embed the magnets

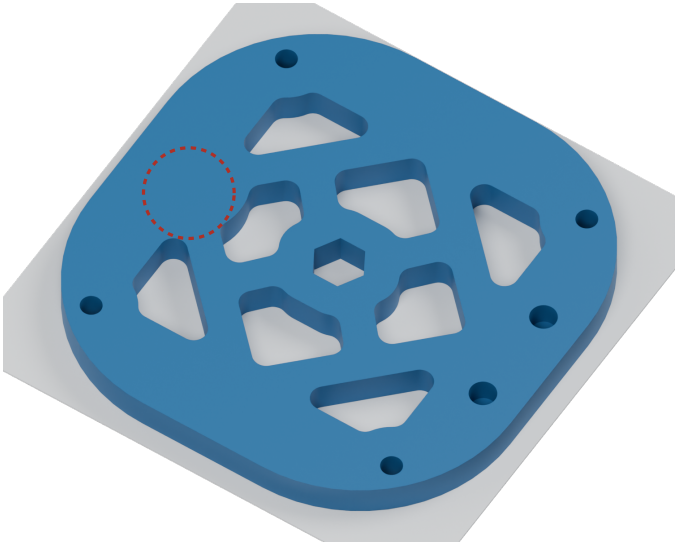
CHECK MAGNET POLARITY

Test with the Hall sensor. The wrong face will be sealed inside.

A / cradle



B / bed base



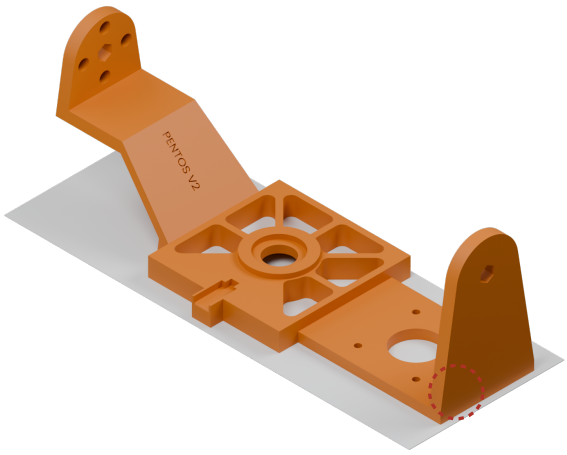
Dashed circle = magnet pocket

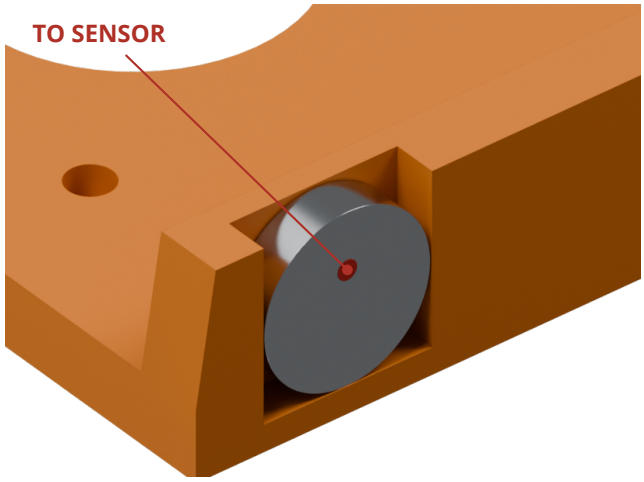
1. **Test each 12 x 3 mm disc magnet with its Hall sensor in the installed orientation.** Confirm the homing input triggers near the magnet and releases when removed. Flip the magnet if needed.
2. **Mark the working face TO SENSOR.** Point it toward the cradle's outer end wall (A), or down toward the print bed (B).
3. **Recheck polarity at the print pause.** Seat the magnet below the next toolpath before resuming.

00A Print the A cradle

PAUSE BEFORE Z 14.40 mm

Object layer 72 / Preview layer 108 (includes supports)



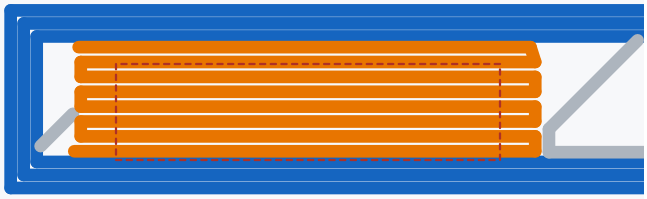


Face toward outer end wall

LAST OPEN: Z 14.20 mm



FIRST COVER: Z 14.40 mm



1. Select your printer and confirm its pause command. At the first-cover layer, add **Insert pause print**.
2. At the pause, seat the magnet below the next toolpath. Resume to enclose it.

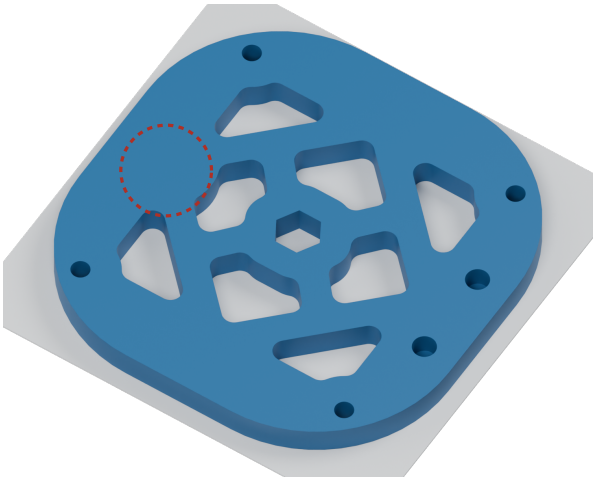
RECHECK POLARITY BEFORE RESUMING.

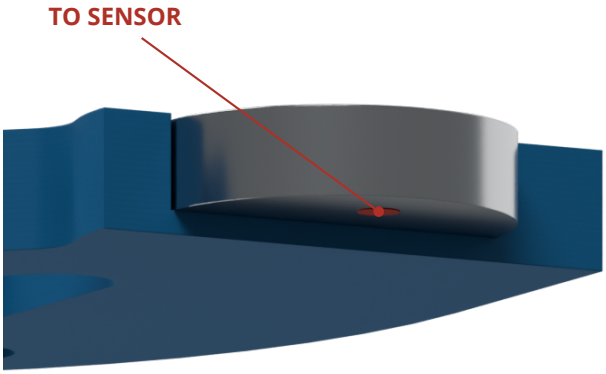
A: marked face toward outer end wall.

00B Print the B base

PAUSE BEFORE Z 4.20 mm

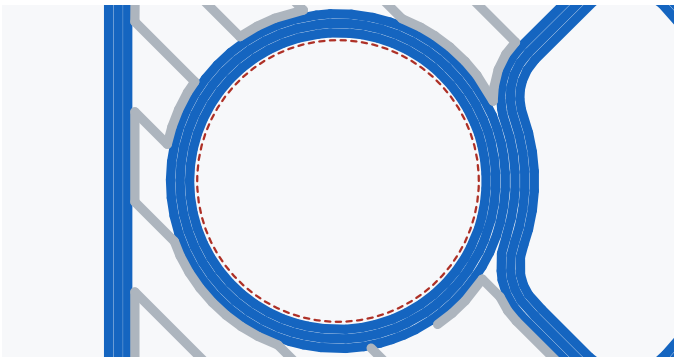
Object layer 21 / Preview layer 21



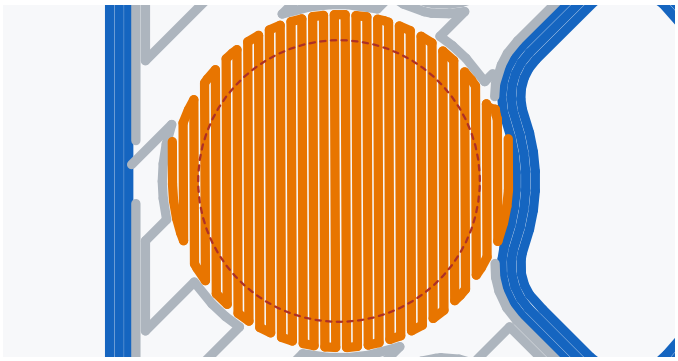


Face DOWN toward the print bed

LAST OPEN: Z 4.00 mm



FIRST COVER: Z 4.20 mm

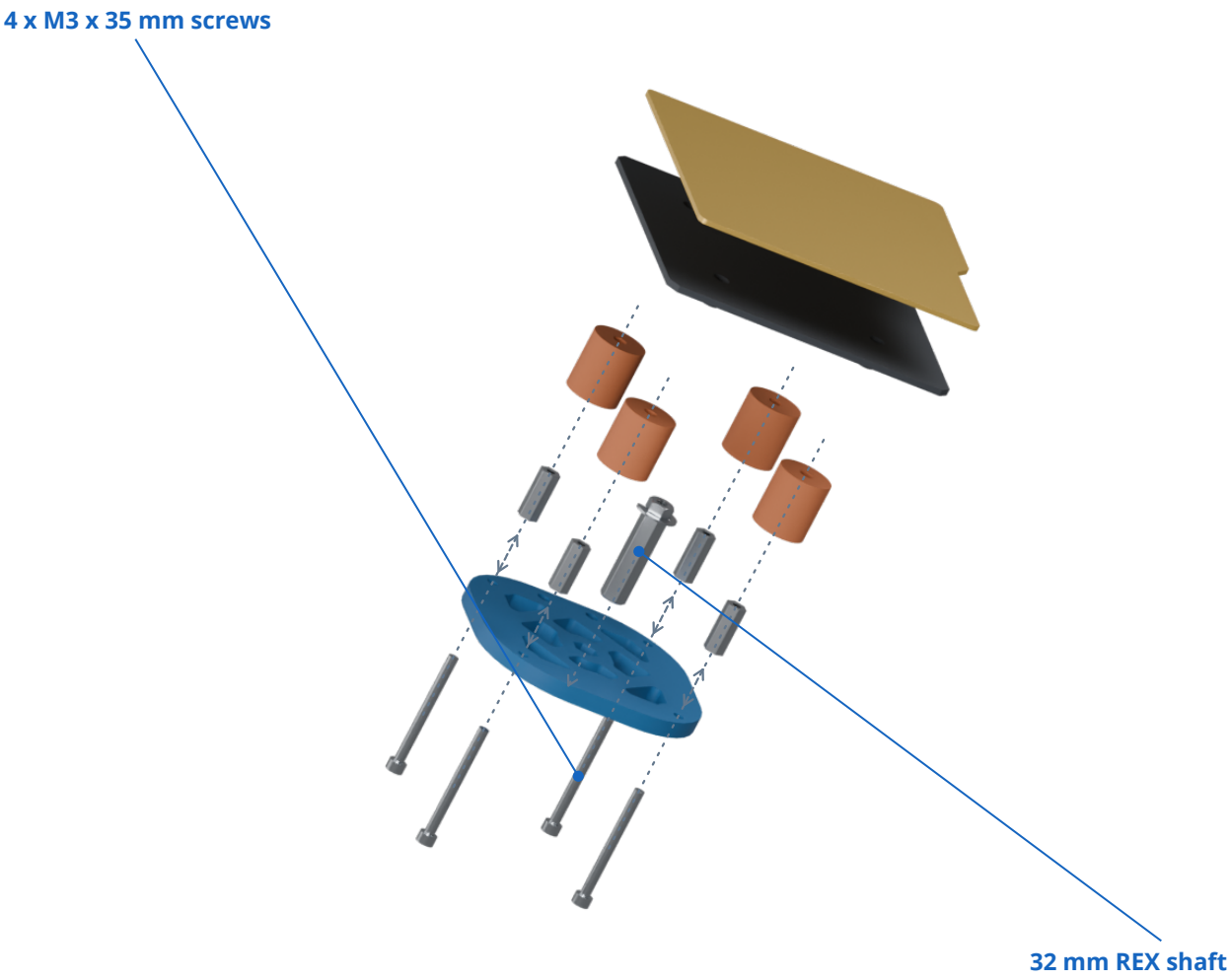


1. Select your printer and confirm its pause command. At the first-cover layer, add **Insert pause print**.
2. At the pause, seat the magnet below the next toolpath. Resume to enclose it.

RECHECK POLARITY BEFORE RESUMING.

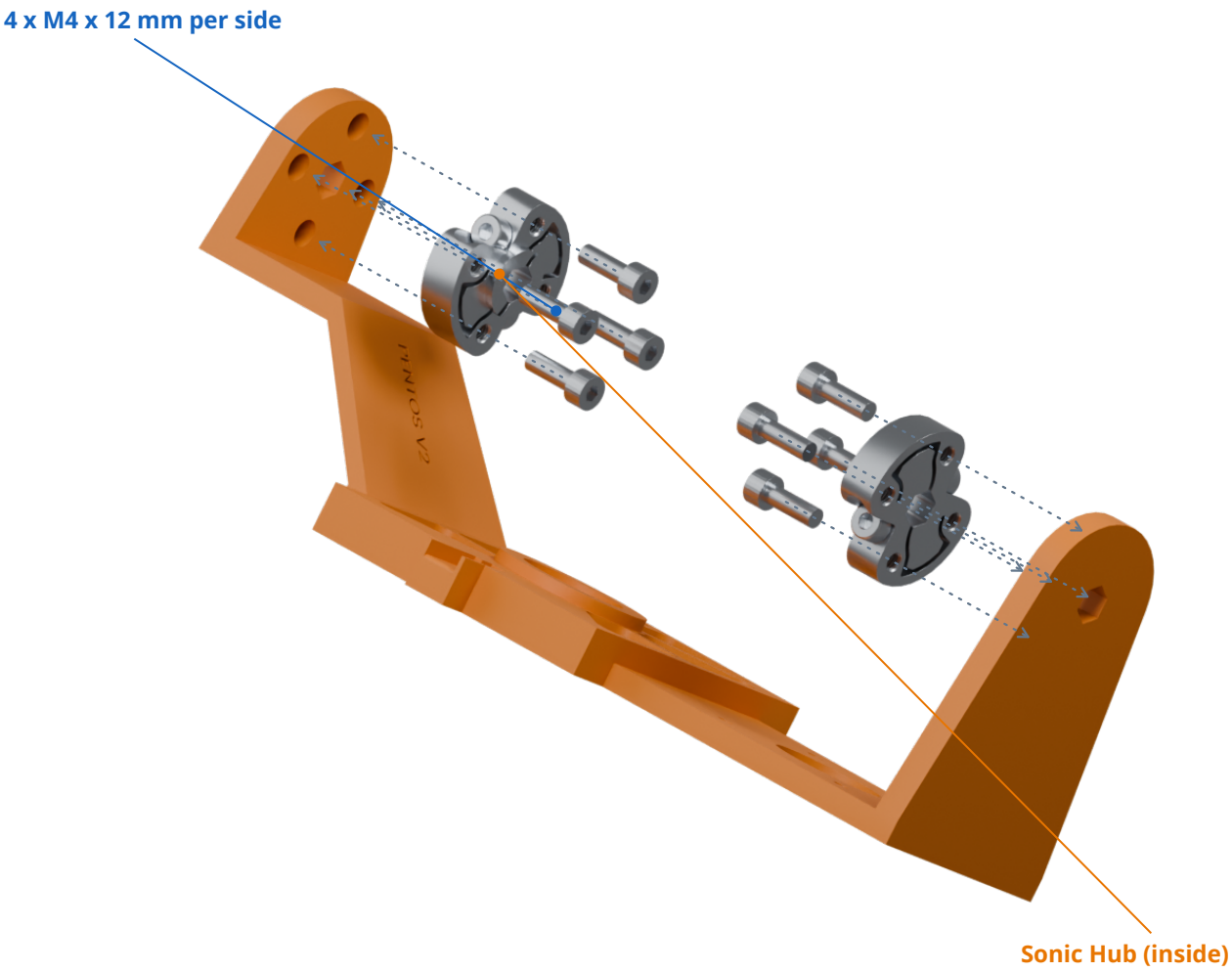
B: marked face DOWN toward the print bed.

01 Assemble the bed



1. Fit four 15 mm standoffs, silicone spacers and the heatbed to the B base.
2. Insert four M3 x 35 mm screws from underneath. Tighten evenly.
3. Add the print surface. Insert the 32 mm REX shaft from the bed side; set the bed aside.

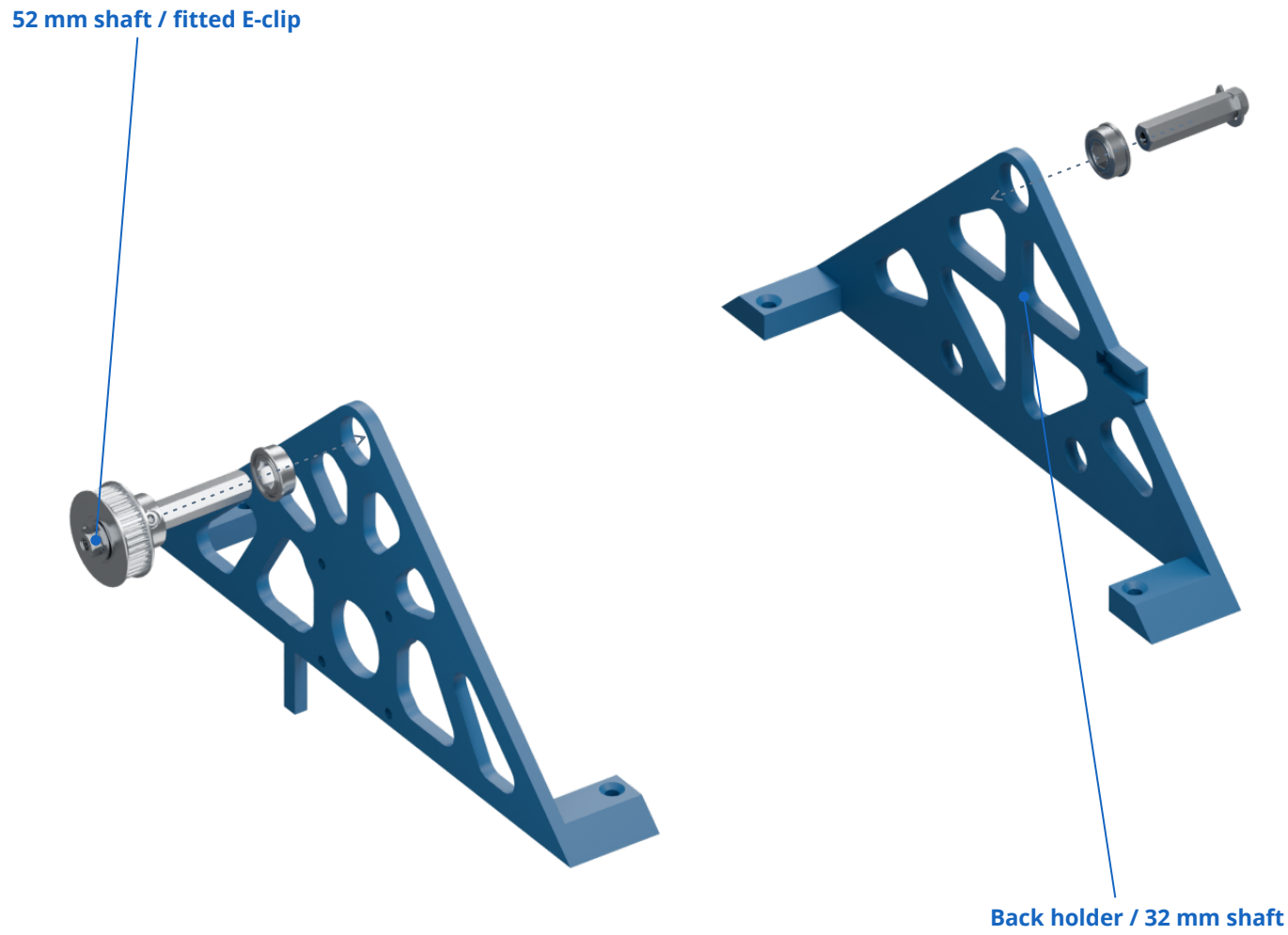
02 Fit the Sonic Hubs



1. Fit both Sonic Hubs inside the cradle.

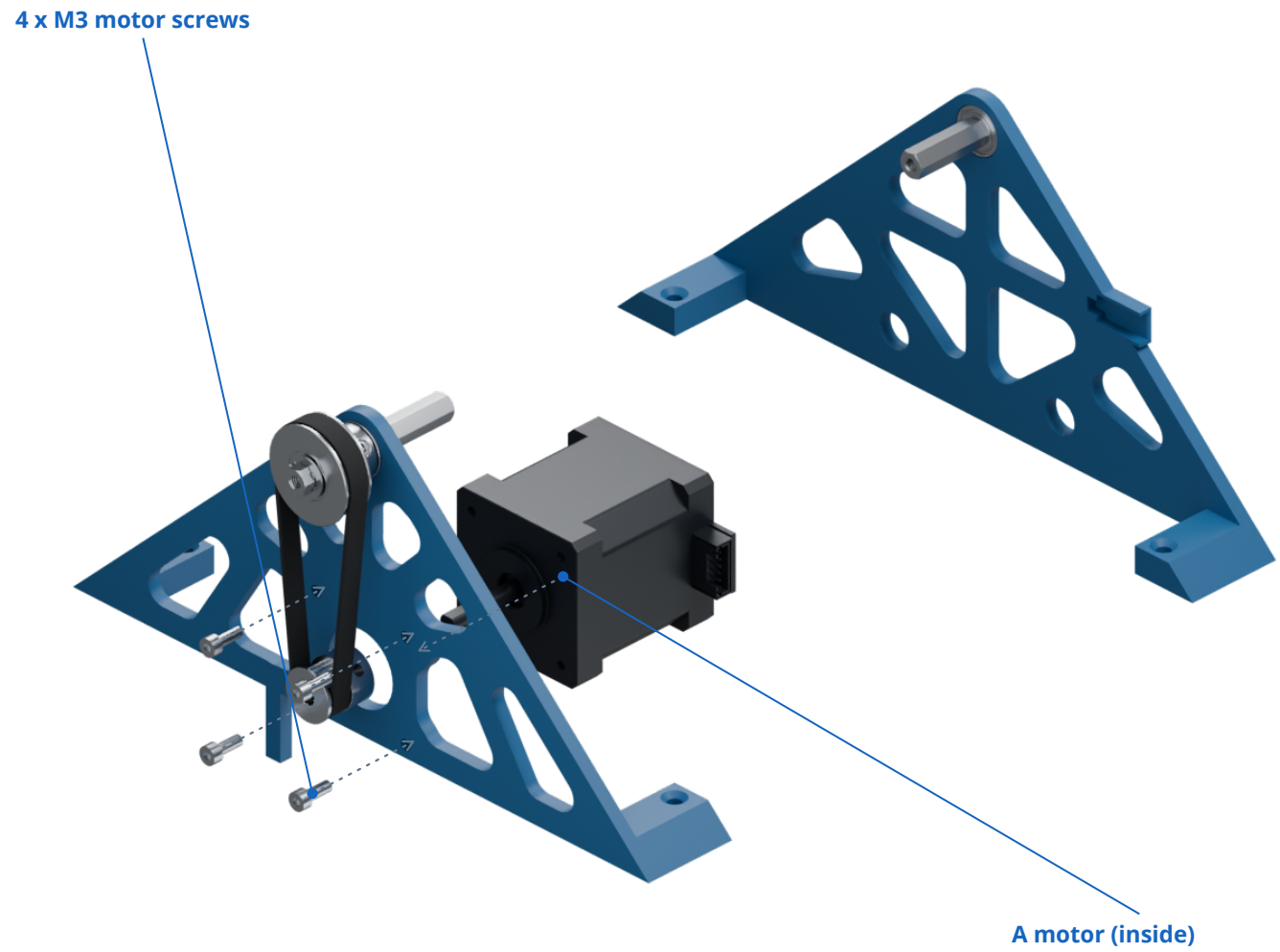
2. Attach each hub with four M4 x 12 mm screws from inside. Leave the shaft clamps loose.

03 Prepare the holders



1. Seat the flanged bearings in both holders.
2. Leave both factory E-clips attached. Slide the 40T pulley onto the 52 mm shaft's free end.
3. Insert each shaft from its clipped side. The clips stay outside the holders.

04 Fit the A-axis drive

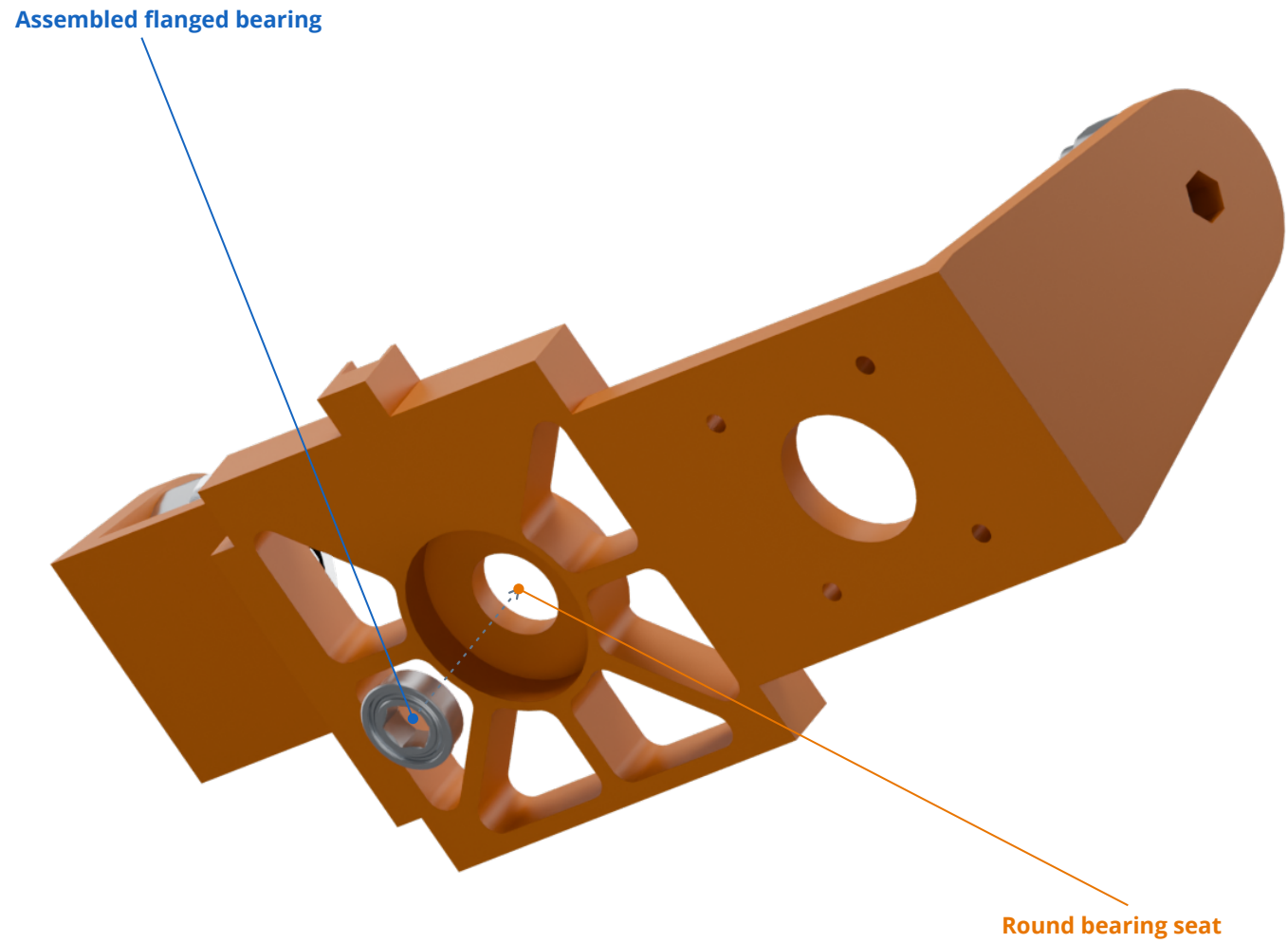


1. Mount the A motor inside the front holder with four M3 screws from outside.

2. Fit the 20T pulley and belt. Align both pulleys and tighten their grub screws.

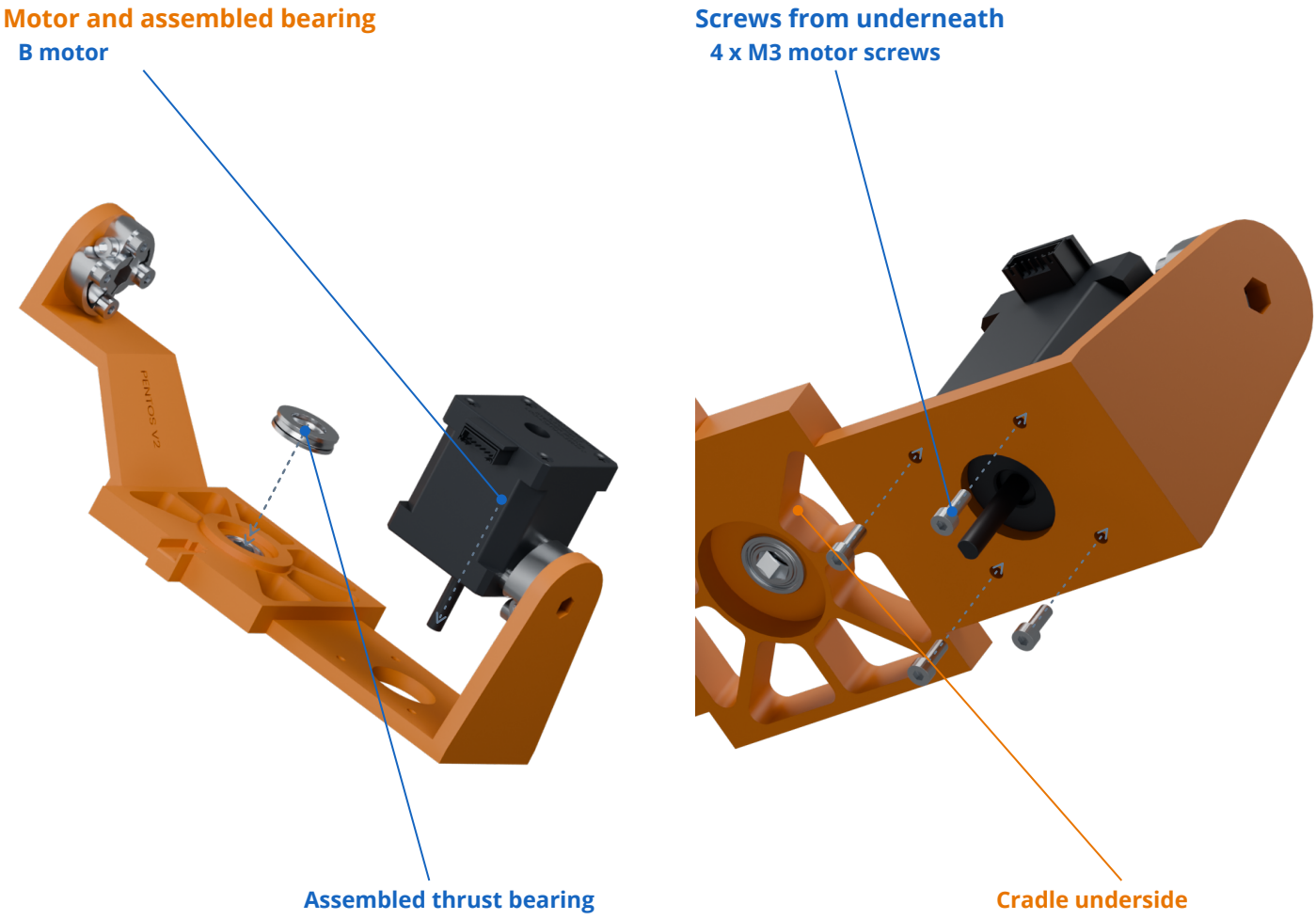
Belts: GT2, 6 mm wide, 184 mm pitch length (92 teeth).

05 Fit the cradle bearing



1. Insert the assembled flanged bearing from underneath the cradle.
2. Seat the flange flat and square against the cradle.

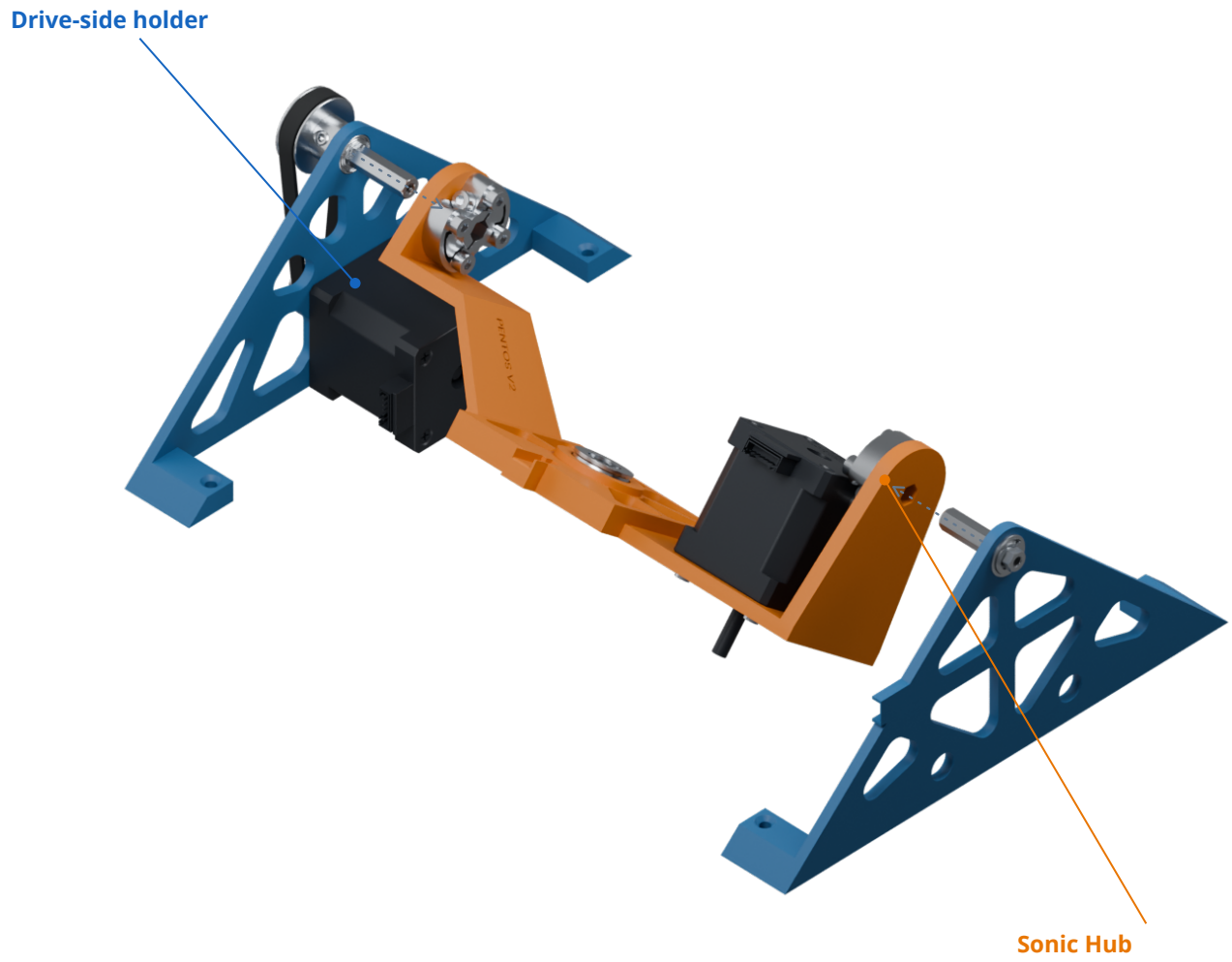
06 Motor and thrust bearing



1. Seat the assembled thrust bearing on the cradle. Keep its stack together.

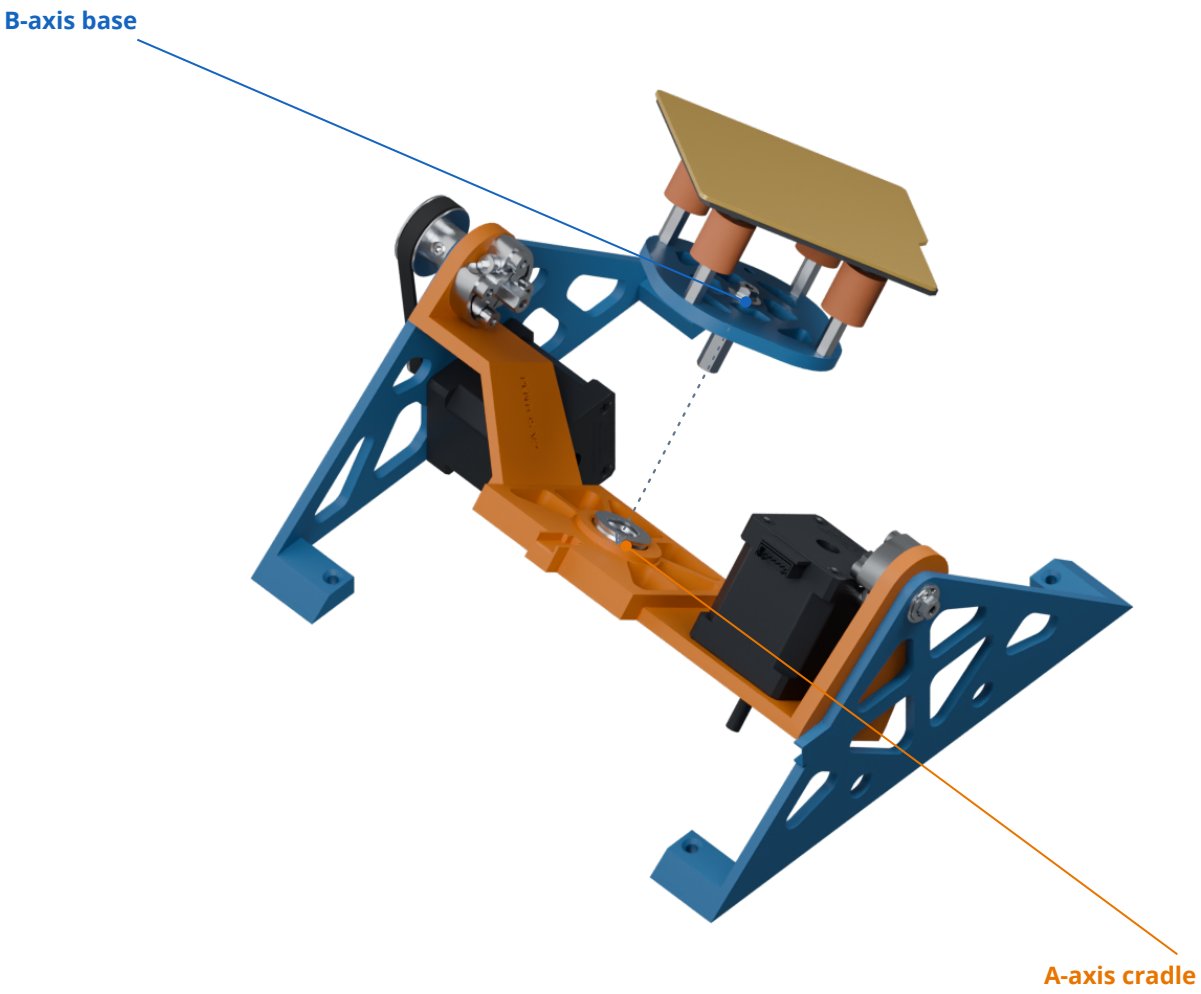
2. Place the B motor on top. Fasten with four M3 screws from underneath.

07 Fit cradle between holders



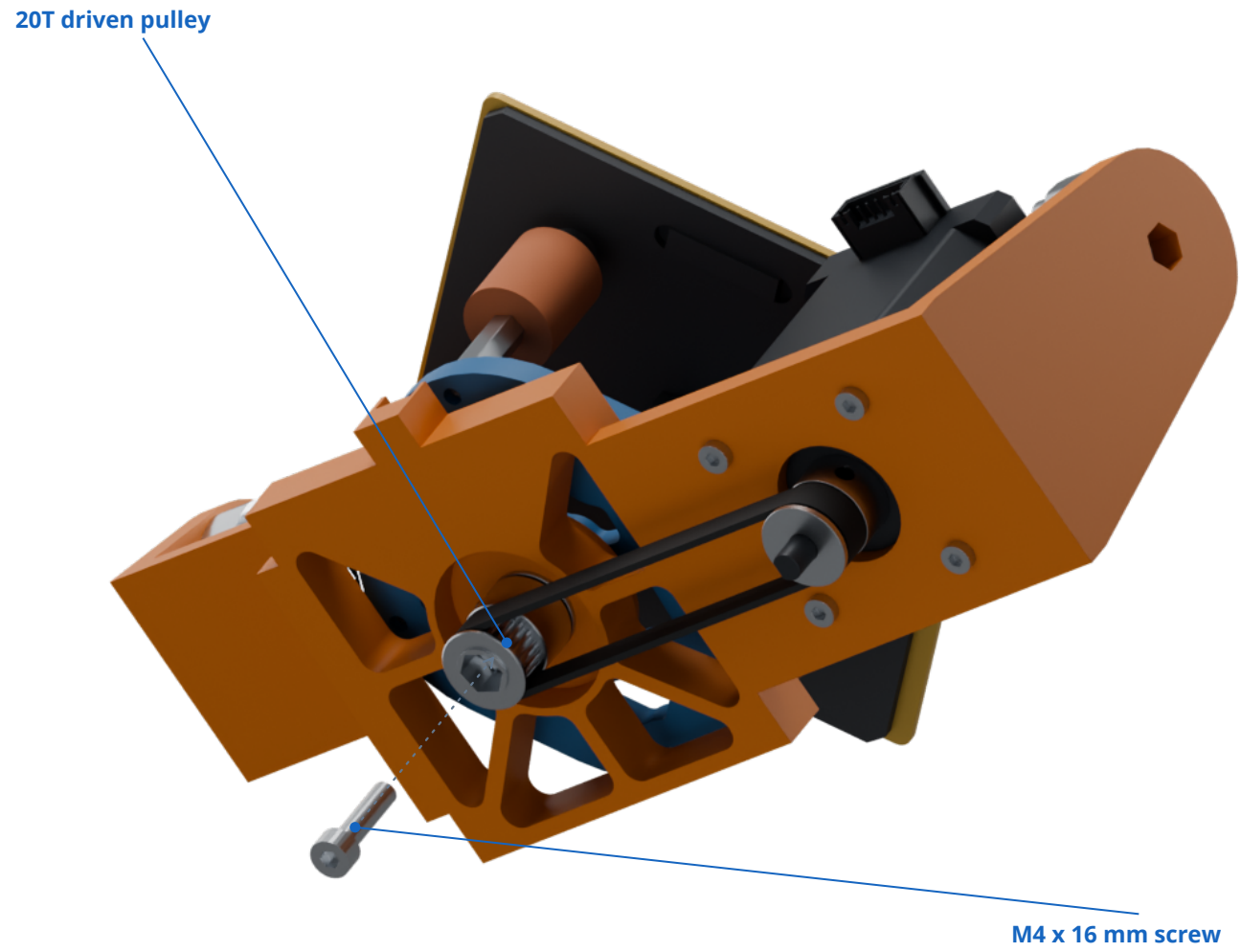
1. Fit the prepared holders to the cradle, sliding their shafts into the Sonic Hubs.
2. Leave the hub clamps loose. Check that the cradle tilts freely.

08 Fit the bed to the cradle



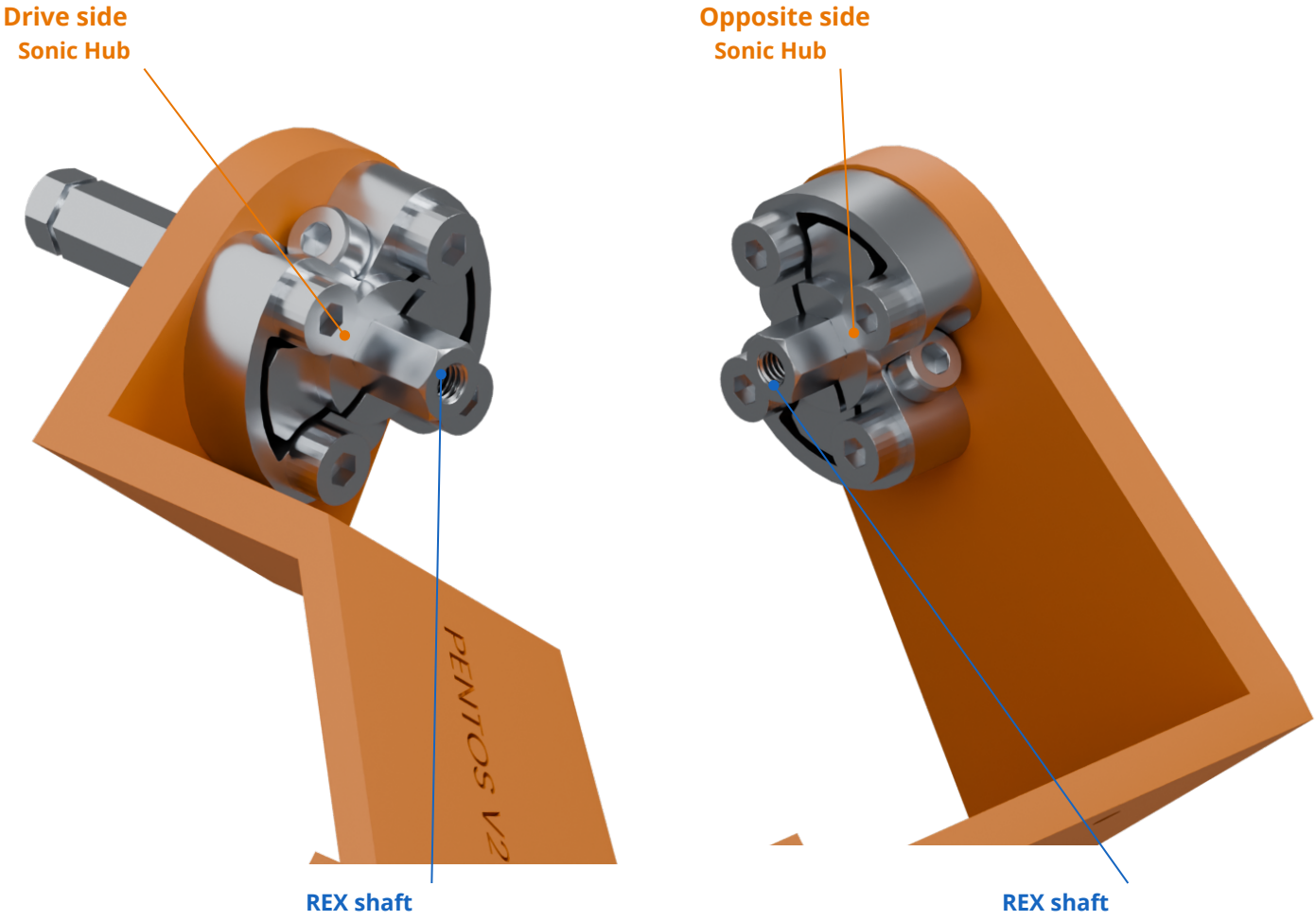
1. Check that both cradle bearings are seated.
2. Lower the bed and guide its shaft through the bearing stack.

09 Install the B drive



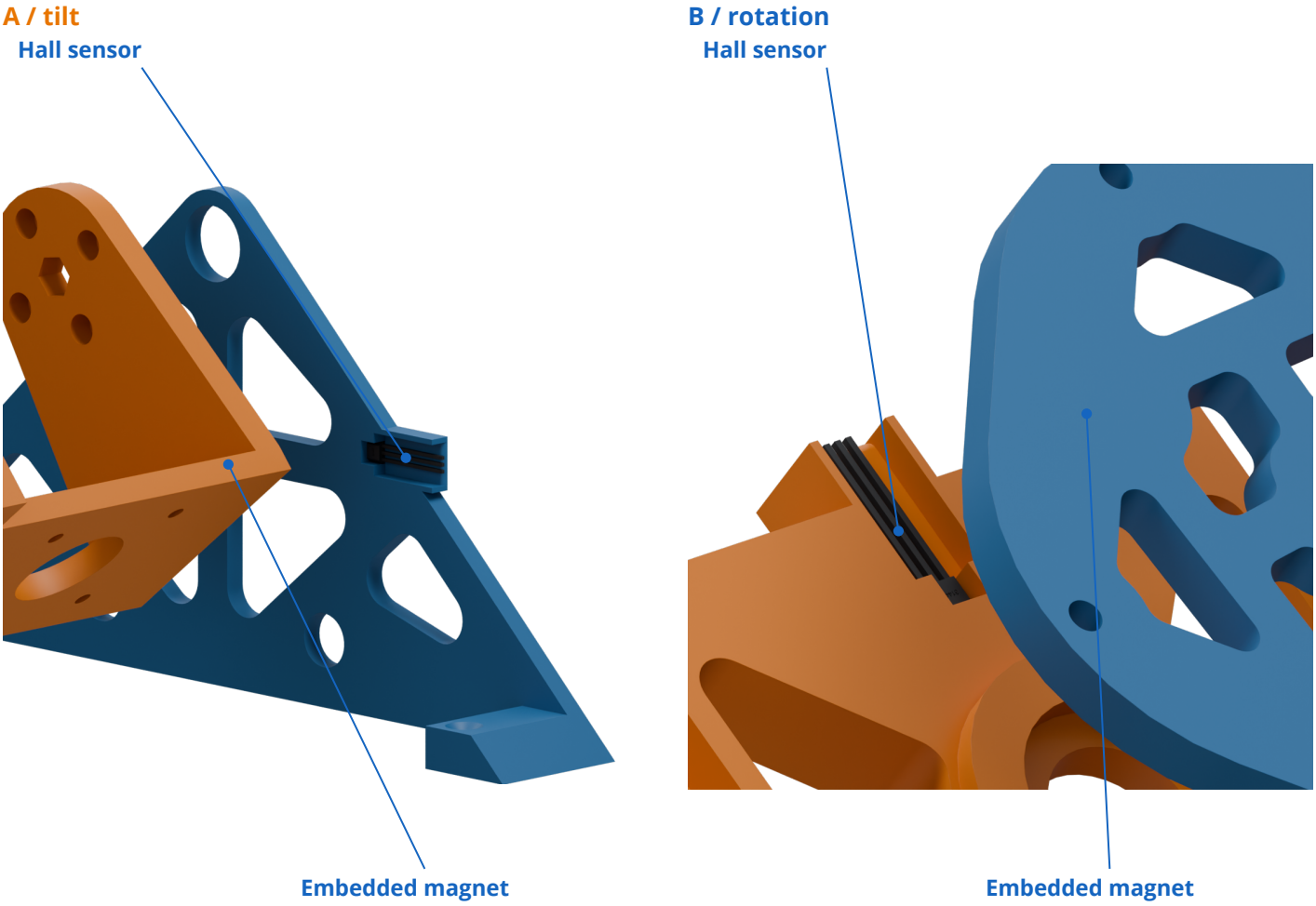
1. Fit both 20T pulleys and the B belt under the cradle.
2. Insert the M4 x 16 mm screw from underneath to secure the pulley and bed.
3. Align the pulleys, tighten the grub screws and check that the bed rotates freely.

10 Align and tighten



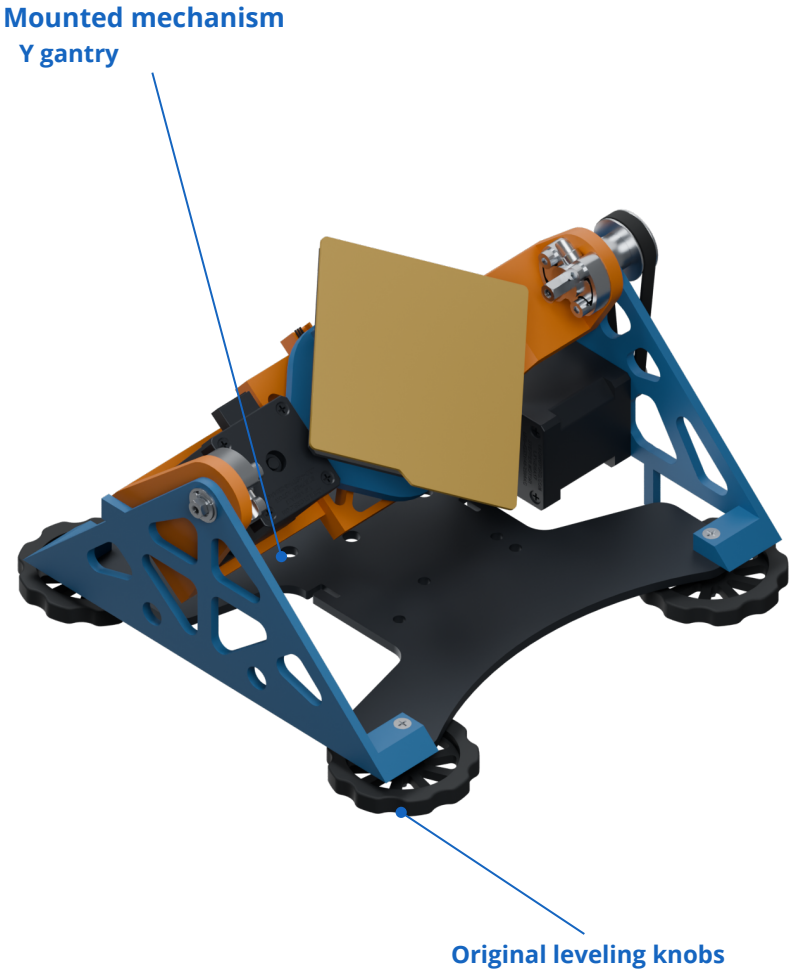
1. Center the cradle and square the shafts.
2. Fit any remaining hub screws. Tighten both hub clamps and all mounting screws.
3. Move A and B by hand; check for binding and belt misalignment.

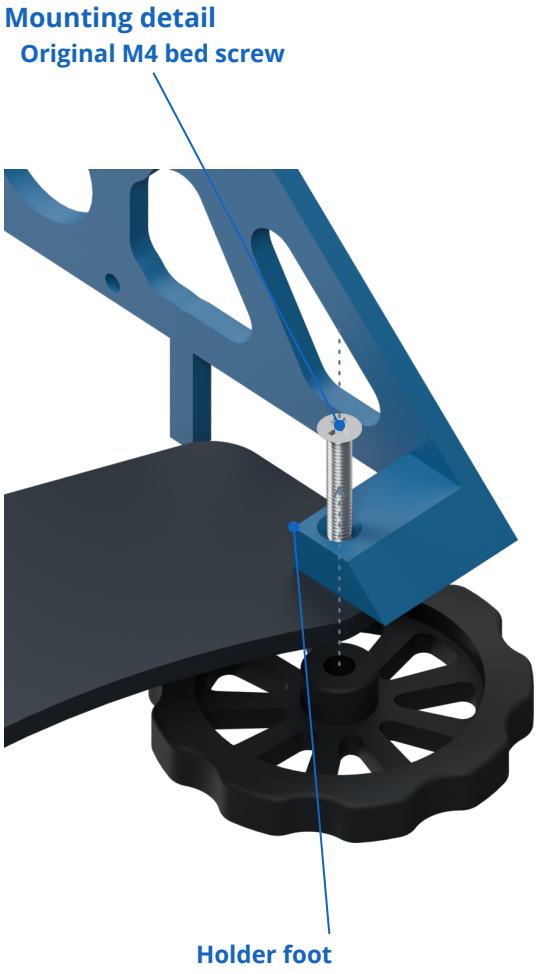
11 Mount the Hall sensors



1. Secure the Hall sensors beside the embedded magnets with hot glue.
2. Route wires clear of the belts, with slack for both axes.
3. Check both sensor signals before homing. Limit B travel to prevent cable twisting.

12 Mount on the Y gantry





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1. Seat the four holder feet directly on the Y gantry.
2. Insert the original M4 countersunk bed screws from above, through the feet and gantry.
3. Refit the leveling knobs underneath and tighten evenly. Check that the full Y travel is clear.